

BCH 301

History and Discovery of Enzymes

Enzymes may be defined as biocatalysts synthesized by living cells that accelerate chemical reactions in living organisms. They are protein in nature (exception— RNA acting as ribozyme), colloidal and thermolabile in character, and specific in their action.

The history of enzymes spans centuries, beginning with observations of fermentation and digestion, evolving through debates on vitalism, and culminating in the recognition of enzymes as proteins.

In the 17th century, scientists observed natural processes such as fermentation—the conversion of sugar to alcohol—and digestion, but the agents responsible for these phenomena remained unknown. In 1665, Robert Hooke described plant cells, laying the foundation for cellular studies and advancing the understanding of life at a microscopic level. Later, in the late 18th century, Antoine Lavoisier investigated fermentation and proposed that chemical reactions occur within living organisms, introducing the idea that life processes have a chemical basis and paving the way for modern biochemistry. Despite these advances, early scientists adhered to the doctrine of **Vitalism**, which held that processes like fermentation and digestion required a special “vital force” present only in living cells. Louis Pasteur reinforced this view in 1857, asserting that fermentation was inseparable from life. This belief dominated scientific thought until the late 19th century, when experiments revealed that enzymes could act outside living cells.

The first major breakthrough came in 1833 when Anselme Payen and Jean-François Persoz isolated **diastase** (now known as amylase) from malt, marking the first enzyme ever discovered and demonstrating its ability to break down starch into sugars. This discovery signaled the beginning of enzymology, even though the true nature of enzymes remained unknown. In 1878, Wilhelm Kühne introduced the term “**enzyme**,” derived from the Greek *en zyme* meaning “in yeast,” reflecting its association with fermentation. The decisive blow to Vitalism occurred in 1897 when Eduard Buchner demonstrated that cell-free yeast extracts could ferment sugar into alcohol. This experiment proved that fermentation was driven by chemical substances—enzymes—rather than a mystical vital force. Buchner’s work earned him the Nobel Prize in Chemistry in 1907 and marked the birth of enzymology as a scientific discipline.

The 20th century brought transformative advances. In 1926, James B. Sumner crystallized urease, conclusively proving that enzymes are proteins—a revolutionary finding that earned him the Nobel Prize in 1946. Soon after, Northrop and Stanley crystallized other enzymes, confirming their protein nature. These breakthroughs were followed by discoveries of coenzymes, enzyme kinetics, and detailed enzyme mechanisms, laying the foundation for modern biochemistry and molecular biology. From early observations of fermentation to the crystallization of enzymes, this journey reflects the transition from mystical explanations to a mechanistic understanding of life processes, shaping the science we know today.

Enzyme Classification and Nomenclature

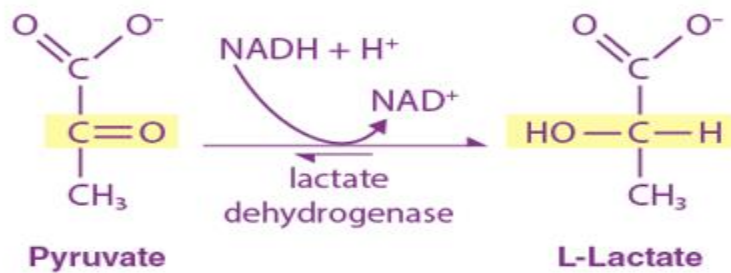
Enzymes are sometimes considered under two broad categories: (a) Intracellular enzymes – They are functional within cells where they are synthesized. (b) Extracellular enzymes – These enzymes are active outside the cell; all the digestive enzymes belong to this group.

Each enzyme is assigned two names. The first is its short, recommended name, convenient for everyday use. In the early days, enzymes were given names by their discoverers in an arbitrary manner. For example, the names pepsin, trypsin and chymotrypsin convey no information about the function of the enzyme or the nature of the substrate on which they act. Sometimes, the suffix-ase was added to the substrate for naming the enzymes e.g. lipase acts on lipids; nuclease on nucleic acids; lactase on lactose. These are known as trivial names of the enzymes which, however, fail to give complete information of enzyme reaction (type of reaction, cofactor requirement etc.).

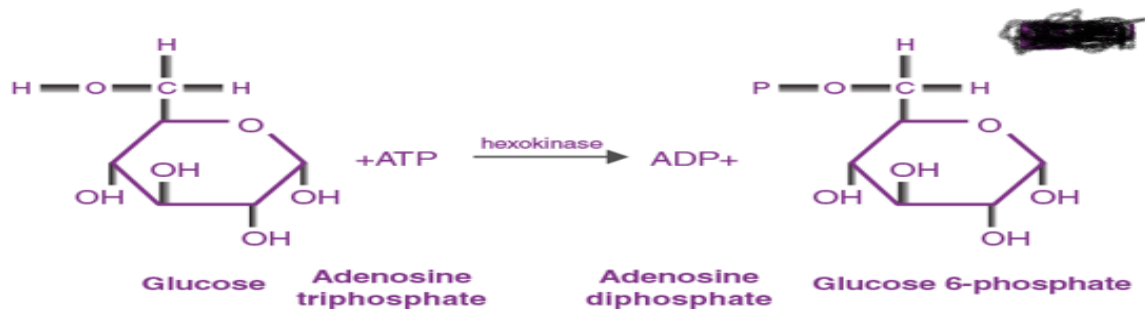
The second is the more complete systematic name, which is used when an enzyme must be identified without ambiguity. The International Union of Biochemistry (IUB) appointed an Enzyme Commission in 1961. This committee made a thorough study of the existing enzymes and devised some basic principles for the classification and nomenclature of enzymes. Since 1964, the IUB system of enzyme classification has been in force. Enzymes are divided into six major classes. Each class on its own represents the general type of reaction brought about by the enzymes of that class. Each class in turn is subdivided into many sub-classes which are further divided. A four digit Enzyme Commission (E.C.) number is assigned to each enzyme representing the class (first digit), sub-class (second digit), sub-sub class (third digit) and the individual enzyme (fourth digit). Each enzyme is given a specific name indicating the substrate, coenzyme (if any) and the type of the reaction catalysed by the enzyme. Although the IUB names for the enzymes are specific and unambiguous, they have not been accepted for general use as they are complex and cumbersome to remember. Therefore, the trivial names, along with the E.C. numbers as and when needed, are commonly used and widely accepted.

E.C.1. Oxidoreductases: Enzymes involved in oxidation-reduction reactions.

They catalyze the transfer of oxygen or hydrogen atoms or electrons from one substrate to another substrate. It is also known as oxidases, reductases or dehydrogenases. An electron donor or an acceptor is also required to complete the reaction since these are redox reactions

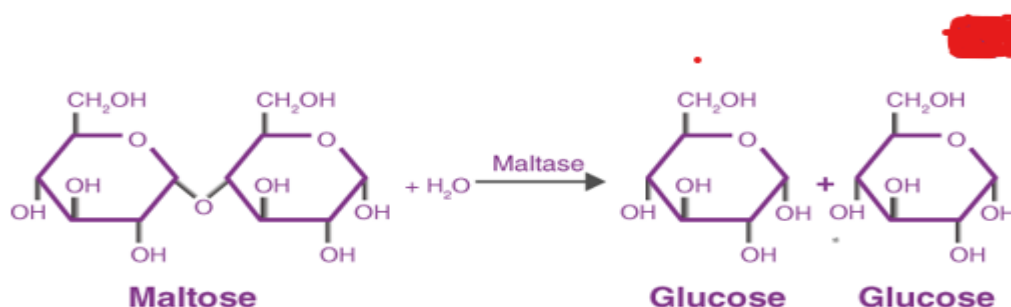


E.C.2. Transferases: Enzymes that catalyse the transfer of functional groups. They includes kinases, acyltransferase, methyltransferase, transketolases, transaldolases, peptidyltransferase etc.



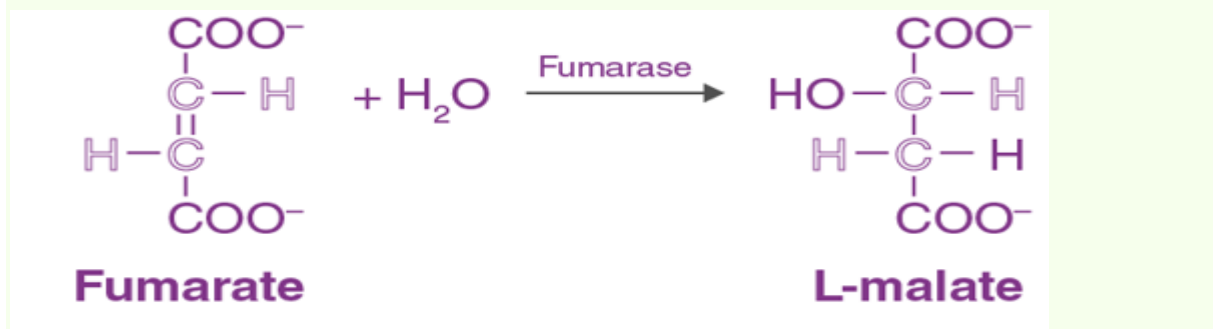
E.C.3. Hydrolases: Enzymes that bring about hydrolysis of various compounds.

Hydrolysis reactions catalyzed by hydrolases commonly break a chemical bond in order to divide a large molecule into two smaller ones. Some common examples of hydrolase enzymes are esterases including lipases, phosphatases, glycosidases, peptidases, and nucleosidases.

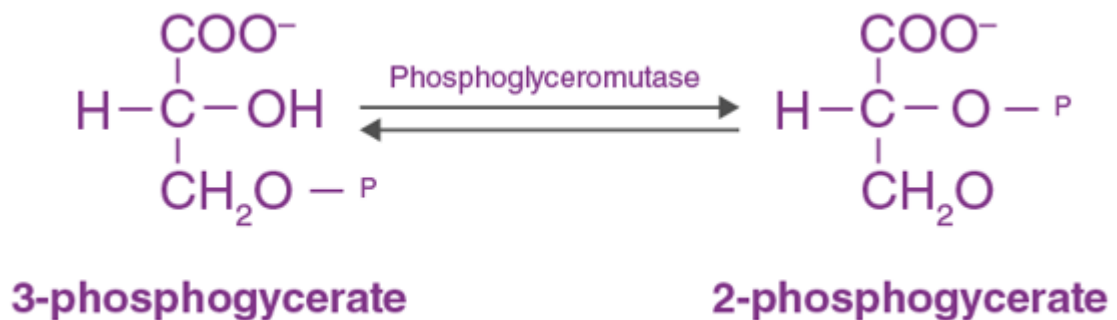


E.C.4. Lyases: Enzymes that specialized in the addition or removal of water, ammonia, CO₂ etc.

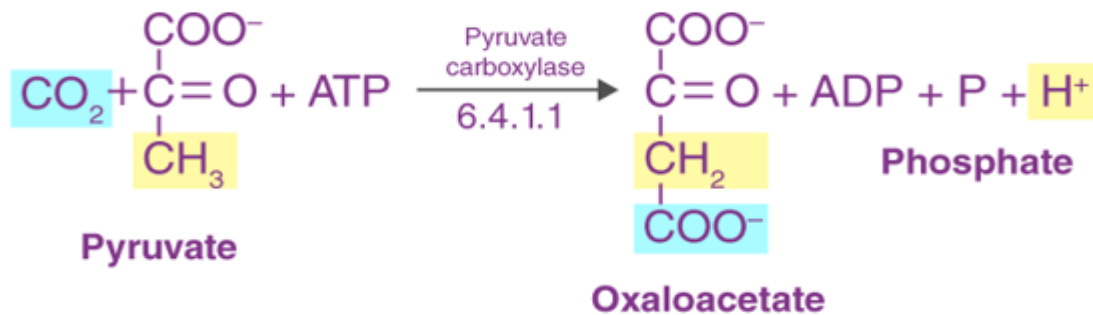
It catalyzes the non-hydrolytic removal of functional groups from substrates. This produces a double bond in the product or a reverse reaction. It includes aldolases and decarboxylases in the removal direction and synthases in the addition direction. Lyases are enzymes that catalyze the elimination of groups from the substrates by processes other than hydrolysis leaving double bonds. It adds water, carbon dioxide or ammonia across double bonds or eliminate these to create double bonds.



E.C.5. Isomerases: Enzymes involved in all the isomerization reactions. Catalyzes isomerization reactions which include cis-trans isomerizations and racemizations.



E.C. 6. Ligases: Enzymes catalyzing the synthetic reactions (Greek : ligate—to bind) where two molecules are joined together and ATP is used.



[The word OTHLIL (first letter in each class) may be memorised to remember the six classes of enzymes in the correct order]

Chemical nature of enzymes

All the enzymes are invariably proteins. In recent years, however, a few RNA molecules have been shown to function as enzymes. Each enzyme has its own tertiary structure and specific conformation which is very essential for its catalytic activity. Some enzymes require molecules other than proteins for enzymic activity. The term holoenzyme refers to the active enzyme with its nonprotein component, whereas the enzyme without its nonprotein moiety is termed an apoenzyme and is inactive. If it is a small organic molecule, it is termed a coenzyme. The non-protein, organic, low molecular weight and dialysable substance associated with enzyme function is known as coenzyme. Coenzymes that only transiently associate with the enzyme are called cosubstrates. Cosubstrates dissociate from the enzyme in an altered state. If the nonprotein moiety is permanently associated with the enzyme and returned to its original form, it is called a prosthetic group. Prosthetic groups are distinguished by their tight, stable incorporation into a protein's structure by covalent or noncovalent forces. Examples include pyridoxal phosphate, flavin mononucleotide (FMN), flavin dinucleotide (FAD), thiamin pyrophosphate, biotin, and the metal ions of Co, Cu, Mg, Mn, Se, and Zn. Metals are the most common prosthetic groups. The roughly one-third of all enzymes that contain tightly bound metal ions are termed metalloenzymes. Coenzymes frequently are derived from vitamins. For example, NAD⁺ contains niacin and FAD contains riboflavin. The coenzyme can be separated by dialysis from the enzyme while the prosthetic group cannot be. The word monomeric enzyme is used if it is made up of a single polypeptide e.g. ribonuclease, trypsin. Some of the enzymes which possess more than one polypeptide (subunit) chain are known as oligomeric enzymes e.g. lactate dehydrogenase, aspartate transcarbamoylase etc. There are certain multienzyme complexes possessing specific sites to catalyze different reactions in a sequence.

Only the native intact multienzyme complex is functionally active and not the individual units, if they are separated e.g. pyruvate dehydrogenase, fatty acid synthase, prostaglandin synthase etc. The enzymes exhibit all the general properties of proteins.

Vitamins

Vitamins may be regarded as organic compounds required in the diet in small amounts to perform specific biological functions for normal maintenance of optimum growth and health of the organism.

History and nomenclature

In the beginning of 20th century, it was clearly understood that the diets containing purified carbohydrate, protein, fat and minerals were not adequate to maintain the growth and health of experimental rats, which the natural foods (such as milk) could do. Hopkins coined the term accessory factors to the unknown and essential nutrients present in the natural foods. Funk (1913) isolated an active principle (an amine) from rice polishings and, later in yeast, which could cure beri-beri in pigeons. He coined the term *vitamine* (Greek: *vita*-life) to the accessory factors with a belief that all of them were amines. It was later realized that only few of them are amines. The term *vitamin*, however, is continued without the final letter 'e'. The usage of A, B and C to vitamins was introduced in 1915 by McCollum and Davis. They first felt there were only two vitamins— fat soluble A and water-soluble B (anti-beriberi factor). Soon another water-soluble anti-scurvy factor named vitamin C was described. Vitamin A was later found to possess two components one that prevents night blindness (vitamin A) and another anti-ricket factor named as vitamin D. A fat soluble factor called vitamin E, in the absence of which rats failed to reproduce properly, was discovered. Yet another fat soluble vitamin concerned with coagulation was discovered in mid 1930s. It was named as vitamin K. In the sequence of alphabets it should have been F, but K was preferred to reflect its function (koagulation). As regards the water soluble factors, vitamin C was identified as a pure substance and named as ascorbic acid. Vitamin B was found to be a complex mixture and nomenclature also became complex. B₁ was clearly identified as anti-beri beri factor. Many investigators carried out intensive research between 1920 and 1930 and went on naming them as the water soluble vitamins B₂, B₃, B₄, B₅, B₆, B₇, B₈, B₉, B₁₀, B₁₁ and B₁₂. Some of them were found to be mixtures of already known vitamins. And for this reason, a few members (numbers!) of the B complex series disappeared from the scene. Except for B₁, B₂, B₆ and B₁₂, names are more commonly used for other B-complex vitamins.

Classification of vitamins

There are about 15 vitamins, essential for humans. They are classified as fat soluble (A, D, E and K) and water soluble (C and B-group) vitamins. The B-complex vitamins may be sub-

divided into energy-releasing (B1, B2, B6, biotin etc.) and hematopoietic (folic acid and B12). Most of the water soluble vitamins exert the functions through their respective coenzymes while only one fat soluble vitamin (K) has been identified to function as a coenzyme.

Fat soluble vitamins

The four vitamins, namely vitamin A, D, E, and K are known as fat or lipid soluble. Their availability in the diet, absorption and transport are associated with fat. They are soluble in fats and oils and also the fat solvents (alcohol, acetone etc.). Fat soluble vitamins can be stored in liver and adipose tissue. They are not readily excreted in urine. Excess consumption of these vitamins (particularly A and D) leads to their accumulation and toxic effects.

Water soluble vitamins

The water-soluble vitamins are a heterogenous group of compounds since they differ chemically from each other. The only common character shared by them is their solubility in water. Most of these vitamins are readily excreted in urine and they are not toxic to the body. Water soluble vitamins are not stored in large quantities (except B12). For this reason, they must be continuously supplied in the diet. The water-soluble vitamins form coenzymes that participate in a variety of biochemical reactions, related to either energy generation or hematopoiesis.